

E085

C# .NET Back End Development

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A mobile app for a hotel chain with 300 properties sends PUT /users/42 with the same full profile body three times because of network timeouts. The app's icon is a gold key. What is the user record's state afterward?

- A. Unknown, because PUT is never safe to retry.
- B. The second and third calls fail with 409 Conflict.
- C. Three stacked updates are applied.
- D. The same as if it had been sent once.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Retried POST /orders requests are creating duplicate orders. Which change best prevents duplicates?

- A. Use PATCH, which is always idempotent.
- B. Require an idempotency key and have the server ignore repeats with the same key.
- C. Switch the endpoint to GET.
- D. Increase the client timeout.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A signed-in clerk at a law firm with a downtown office sends DELETE /invoices/9 with a valid token, but her role does not allow deleting invoices. The firm has 45 lawyers. Which status code should the API return?

- A. 500 Internal Server Error
- B. 401 Unauthorized
- C. 403 Forbidden
- D. 400 Bad Request

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An API returns 401 Unauthorized. What most likely caused it?

- A.** The account is permanently banned in every case.
- B.** The user was authenticated but lacked the needed role.
- C.** The requested record does not exist.
- D.** The request had a missing, expired, or invalid token.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A team at a food distributor that serves 2,000 restaurants needs an endpoint that returns the line items for order 57. Their API gateway was upgraded last month. Which route and method follow REST conventions?

- A.** GET /orders/57/items
- B.** GET /order-items-57
- C.** POST /orders/57/items
- D.** GET /getOrderItems?orderId=57

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A controller marked [ApiController] at a ticketing company with a busy summer season accepts a model where Quantity has [Range(1, 100)]. A client posts Quantity = 0. The company's database runs on SQL Server. What happens?

- A.** ASP.NET always returns 422 for validation errors.
- B.** A 500 error from SQL Server.
- C.** The action runs and saves Quantity = 0.
- D.** A 400 response with validation details; the action does not run.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A developer at a fitness-tracking startup with 1 million users registers PriceCache as a singleton. PriceCache takes ApplicationDbContext, registered as scoped, in its constructor. The startup just raised a funding round. What is the problem and the right fix?

- A. A captive dependency; inject IServiceScopeFactory to create a scope when needed, or make PriceCache scoped.
- B. Register ApplicationDbContext as transient.
- C. No problem; singletons can depend on any lifetime.
- D. Register ApplicationDbContext as a singleton too.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which built-in lifetime creates exactly one instance per HTTP request?

- A. Singleton
- B. Scoped
- C. Transient
- D. Per-thread

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

CalculateShipping(weightKg) at a garden-supply retailer with a catalog of 8,000 products accepts weights greater than 0 and up to 50 kg. The code was written in C# 12. Which set of test inputs best covers typical use and edge cases?

- A. 0, 0.01, 25, 50, and 50.01
- B. 10 and 25
- C. 1,000 random values between 1 and 49
- D. One passing value is enough to prove it works.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A payroll API at a staffing firm, which moved to the cloud in 2024, has its production database password inside appsettings.json in the Git repository. The repo has 3,000 commits. What is the right fix?

- A. Move it to appsettings.Production.json and commit that.
- B. Rotate the password and load it from a secret store or environment variable at run time.
- C. Comment it out in the JSON file.
- D. Hard-code it in Program.cs instead.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An operations lead at a ride-share company with a bright pink brand wants to be paged within a minute if the API error rate rises above 2%. The API handles 4,000 requests a second. Which observability tool fits?

- A. Distributed tracing of every request.
- B. A health-check endpoint only.
- C. An error-rate metric with an alert rule.
- D. Reading the logs every morning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A request passes through three services and takes 4 seconds. Which tool best shows which service is using the time?

- A. An error-rate metric
- B. A single log line at the end of the request
- C. Distributed tracing
- D. A health-check endpoint

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A mobile app ~~for a hotel chain with 300 properties~~ sends PUT /users/42 with the same full profile body three times because of network timeouts. ~~The app's icon is a gold key.~~ What is the user record's state afterward?

Target word: effect of repeated PUT **Decides:** PUT replaces the resource with the same body each time.

A	Absolute language	Idempotency is exactly what makes PUT retry-safe.
B	Adjacent category	Replacing with identical data is not a conflict.
C	Confusion set	That describes non-idempotent POST-style behavior.
D	Correct	PUT is idempotent.

2. Answer B Concept 1 • transfer

Retried POST /orders requests are creating duplicate orders. Which change best prevents duplicates?

Target word: reverse: make a non-idempotent call retry-safe **Decides:** POST is not idempotent by default.

A	Absolute language	PATCH is not guaranteed to be idempotent.
B	Correct	The server recognizes retries and returns the original result.
C	Inverted operation	GET must not create or change data.
D	Dropped element	Fewer retries, but duplicates are still possible.

3. Answer C Concept 2

A signed-in clerk at a law firm ~~with a downtown office~~ sends DELETE /invoices/9 with a valid token, but her role does not allow deleting invoices. ~~The firm has 45 lawyers.~~ Which status code should the API return?

Target word: status for known user lacking permission **Decides:** Authenticated, not authorized.

A	Dropped element	Ignores that the refusal is intended, not a server fault.
B	Confusion set	401 means credentials are missing or invalid; hers are valid.
C	Correct	The caller is known but not allowed.
D	Adjacent category	The request is well formed; the problem is permission.

4. Answer D Concept 2 • transfer

An API returns 401 Unauthorized. What most likely caused it?

Target word: reverse: code to cause **Decides:** 401 concerns authentication.

A	Absolute language	401 usually just means credentials must be supplied or refreshed.
B	Inverted operation	That is 403.
C	Adjacent category	That is 404.
D	Correct	401 signals failed or absent authentication.

5. Answer A Concept 3

A team at a food distributor ~~that serves 2,000 restaurants~~ needs an endpoint that returns the line items for order 57. ~~Their API gateway was upgraded last month.~~ Which route and method follow REST conventions?

Target word: REST route choice **Decides:** Read a sub-collection of an order.

A	Correct	Plural nouns, hierarchy, and GET for a read.
B	Confusion set	Flattens the ID into the name and loses the resource hierarchy.
C	Inverted operation	POST creates; this is a read.
D	Adjacent category	RPC style: a verb in the path.

6. Answer D Concept 4

A controller marked [ApiController] at a ticketing company ~~with a busy summer season~~ accepts a model where Quantity has [Range(1, 100)]. A client posts Quantity = 0. ~~The company's database runs on SQL Server.~~ What happens?

Target word: result of invalid model **Decides:** [ApiController] rejects invalid model state automatically.

A	Absolute language	The default is 400.
B	Camouflage	The request never reaches the database.
C	Dropped element	Ignores the [ApiController] validation step.
D	Correct	Automatic model validation returns problem details.

7. Answer A Concept 5

A developer at a fitness-tracking startup ~~with 1 million users~~ registers PriceCache as a singleton. PriceCache takes ApplicationDbContext, registered as scoped, in its constructor. ~~The startup just raised a funding round.~~ What is the problem and the right fix?

Target word: lifetime mismatch **Decides:** A singleton would capture a scoped service for the whole app's life.

A	Correct	Avoids sharing one non-thread-safe DbContext across requests.
B	Adjacent category	The singleton still captures one instance forever.
C	Absolute language	Singletons must not capture scoped services.
D	Inverted operation	Shares one DbContext across all requests, making it worse.

8. Answer B Concept 5 · transfer

Which built-in lifetime creates exactly one instance per HTTP request?

Target word: name the lifetime **Decides:** One per request = scoped.

A	Inverted operation	One instance for the whole application.
B	Correct	A scope is created for each request.
C	Confusion set	New instance every time it is requested, even within one request.
D	Adjacent category	Not a built-in ASP.NET Core lifetime.

9. Answer A Concept 6

CalculateShipping(weightKg) at a garden-supply retailer ~~with a catalog of 8,000 products~~ accepts weights greater than 0 and up to 50 kg. ~~The code was written in C# 12.~~ Which set of test inputs best covers typical use and edge cases?

Target word: test case selection **Decides:** Boundaries are 0 (excluded) and 50 (included).

A	Correct	Hits both boundaries, just inside and outside, plus a typical value.
B	Dropped element	Skips every boundary.
C	Adjacent category	Many cases, but none at the edges.
D	Absolute language	A single case cannot show boundary behavior.

10. Answer B Concept 7

A payroll API at a staffing firm, ~~which moved to the cloud in 2024,~~ has its production database password inside appsettings.json in the Git repository. ~~The repo has 3,000 commits.~~ What is the right fix?

Target word: secret handling **Decides:** A secret is stored in a committed file.

A	Confusion set	Environment files are for settings, not secrets; it is still in source.
B	Correct	Removes the secret from source and invalidates the exposed one.
C	Dropped element	The secret is still visible in the repository and its history.
D	Inverted operation	Still in source, and harder to change.

11. Answer C Concept 8

An operations lead at a ride-share company ~~with a bright pink brand~~ wants to be paged within a minute if the API error rate rises above 2%. The API handles 4,000 requests a second. Which observability tool fits?

Target word: tool for alerting on a rate **Decides:** A rate threshold needs a metric with an alert.

A	Confusion set	Traces show where time goes on a request; metrics drive rate alerts.
B	Adjacent category	Reports up or down, not an error rate.
C	Correct	Metrics aggregate rates cheaply and drive alerts.
D	Dropped element	Misses the one-minute paging need.

12. Answer C Concept 8 · transfer

A request passes through three services and takes 4 seconds. Which tool best shows which service is using the time?

Target word: reverse: question to tool **Decides:** Time breakdown across services = tracing.

A	Confusion set	Shows how often errors happen, not where time goes.
B	Dropped element	Records the total time, not the breakdown.
C	Correct	Spans show time spent in each service on one request.
D	Adjacent category	Only reports whether a service is up.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	HTTP methods and idempotency	GET is safe; PUT and DELETE are idempotent (repeating them leaves the same state); POST is not, so retries need an idempotency key.	Assuming any retried request is harmless, or that repeated PUTs stack changes.	A client retry after a timeout.
2	Status codes: 401 vs. 403 and friends	401 = not authenticated (missing or invalid credentials); 403 = authenticated but not allowed; 400 = bad input; 404 = not found; 5xx = server fault.	Returning 401 when the user is known but lacks permission.	Who the caller is versus what they are allowed to do.
3	RESTful resource design	URLs name resources with plural nouns and hierarchy (/orders/57/items); the HTTP method carries the action.	Putting verbs in the path (/getOrderItems) or using POST to read.	Choosing a route for a read or write operation.
4	Model validation at the boundary	With [ApiController], invalid model state automatically returns 400 with validation details before the action runs.	Expecting invalid input to reach the database or produce a 500.	Data annotations such as [Required] or [Range] and an out-of-range value.
5	Dependency injection lifetimes	Transient = new each time; scoped = one per request; singleton = one for the app. A singleton must not capture a scoped service such as a DbContext.	Injecting a scoped DbContext into a singleton, or fixing it by making the DbContext singleton.	A service registration lifetime and its constructor dependencies.
6	Unit tests for typical and edge cases	Test values at, just inside, and just outside each boundary plus typical values; one passing case proves little.	Testing only comfortable middle values.	A method with a valid range and a question about which tests to write.
7	Environment configuration and secrets	Keep environment settings in layered configuration (appsettings.{Environment}.json, environment variables) and secrets in user secrets or a vault, never in committed files.	Moving a secret to a different committed JSON file and calling it secure.	A connection string or key in appsettings.json.
8	Logs, metrics, and traces	Metrics show rates and trends for alerting; logs record event detail; distributed traces show where time is spent across services.	Using logs alone for real-time alerting, or metrics to find which downstream call is slow.	“Alert when...” vs. “which service is slow...” vs. “what happened on this request.”

Course scope (public description): This course is the third in a three-course C# sequence in the BSAIE program (AI Engineering with C#, Advanced C#, and C# .NET Backend Development) and focuses on the development of enterprise-ready backend services using ASP.NET. Learners extend their C# proficiency into professional software engineering by designing and implementing RESTful APIs that meet both functional and non-functional requirements, including scalability, security, and maintainability. The course emphasizes validation of service behavior through comprehensive automated unit testing that addresses typical usage scenarios and edge cases. Learners also configure .NET applications for deployment, incorporating logging, monitoring, and metrics to support operational observability and ongoing service reliability. Through applied backend engineering practices, learners build the skills necessary to deploy and operate production-grade services and prepare for integrating AI

models into real-world systems that require scalable, secure, and observable backend infrastructure.